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import numpy as np

def cross_entropy_loss(y_true, y_pred):
    y_pred_clipped = np.clip(y_pred, 1e-15, 1 - 1e-15)
    print("Clipped predicted probabilities:\n", y_pred_clipped)
    print("\n\n")

    log_y_pred = np.log2(y_pred_clipped)
    print("Log base 2 of clipped predicted probabilities:\n", log_y_pred)
    print("\n\n")

    y_true_log_y_pred = y_true * log_y_pred
    print("Element-wise product of true labels and log base 2 of predicted probabilities:\n", y_true_log_y_pred)
    print("\n\n")

    loss = -np.sum(y_true_log_y_pred) / y_true.shape[0]
    print("Summation of Result: ", -np.sum(y_true_log_y_pred))

    return loss

if __name__ == "__main__":
    y_true = np.array([[0, 0, 0, 1],
                       [0, 1, 0, 0],
                       [0, 0, 1, 0]])

    y_pred = np.array([[0.03, 0.07, 0.07, 0.82],
                       [0.85, 0.04, 0.04, 0.07],
                       [0.07, 0.04, 0.85, 0.04]])

    loss = cross_entropy_loss(y_true, y_pred)
    print(f"\n\nCross entropy loss (log base 2): {loss}")

```



Clipped predicted probabilities:

```

[[0.03 0.07 0.07 0.82]
 [0.85 0.04 0.04 0.07]
 [0.07 0.04 0.85 0.04]]

```

Log base 2 of clipped predicted probabilities:

```

[[-5.05889369 -3.83650127 -3.83650127 -0.28630419]
 [-0.23446525 -4.64385619 -4.64385619 -3.83650127]
 [-3.83650127 -4.64385619 -0.23446525 -4.64385619]]

```

Element-wise product of true labels and log base 2 of predicted probabilities:

```

[[-0.          -0.          -0.          -0.28630419]
 [-0.         -4.64385619 -0.          -0.          ]
 [-0.          -0.          -0.23446525 -0.          ]]

```

Summation of Result: 5.1646256285683885

Cross entropy loss (log base 2): 1.7215418761894627